Chapter 1 : Mathematical Modeling and Problem Solving

Numerical methods are techniques by which mathematical problems are formulated using arithmetic operations.

Noncomputer methods

1) Analytical (exact) method : useful for a limited number of problems

2) Graphical Solutions : e.g. plots. Limited to problems in 3D or fewer dimensions.

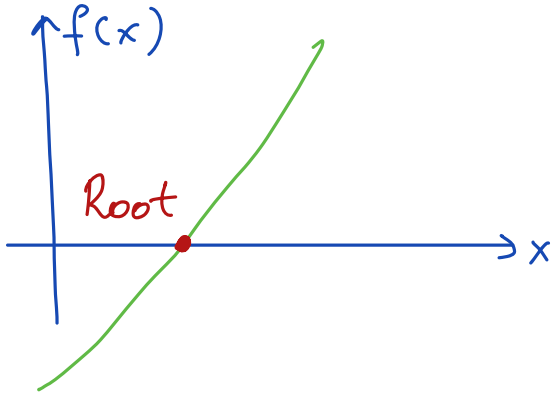
Why study numerical methods?

- NM are very good problem-solving tools.
- use computer programs
- Learn to program in MATLAB :
- Acknowledge and control errors of approximations.
- NM reinforce your understanding of math.

Subject Areas to be Covered :

1) Roots of Equations

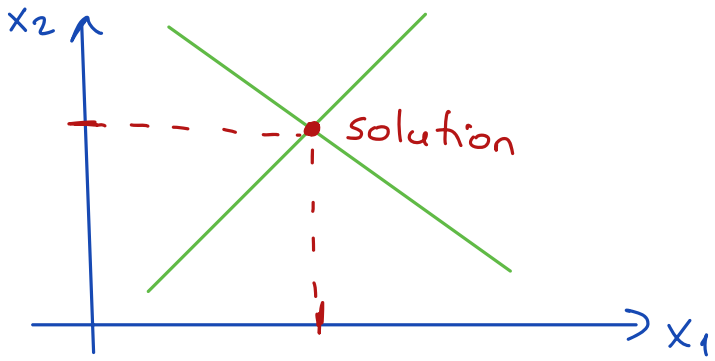
Solve $f(x)=0$ for x



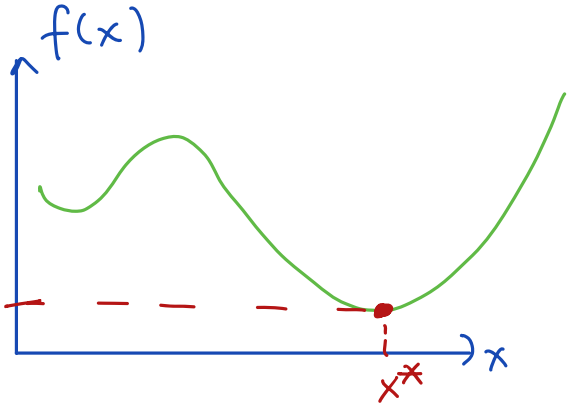
2) Systems of Linear Algebraic Equations

$$a_{11}x_1 + a_{12}x_2 = c_1$$
$$a_{21}x_1 + a_{22}x_2 = c_2$$

Given a's, c's
Solve x's.



3) Optimization

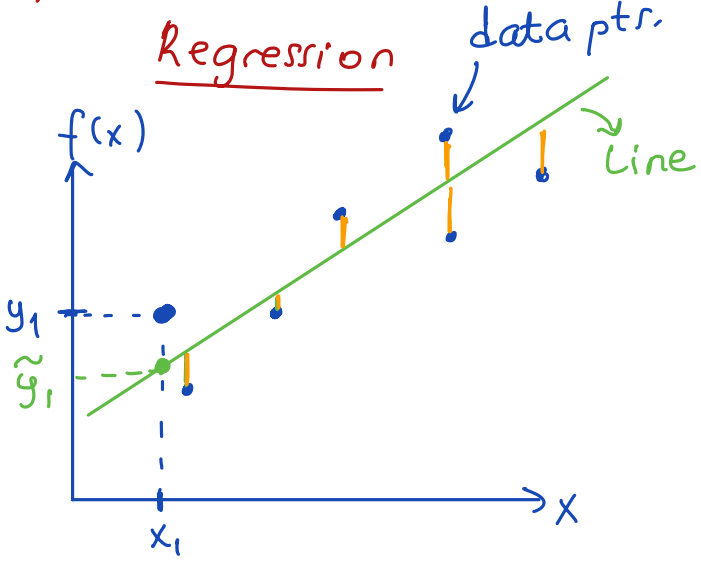


Determine x that gives optimum $f(x)$

- single and multi-variable unconstrained optimization
- constrained optimization (linear programming)

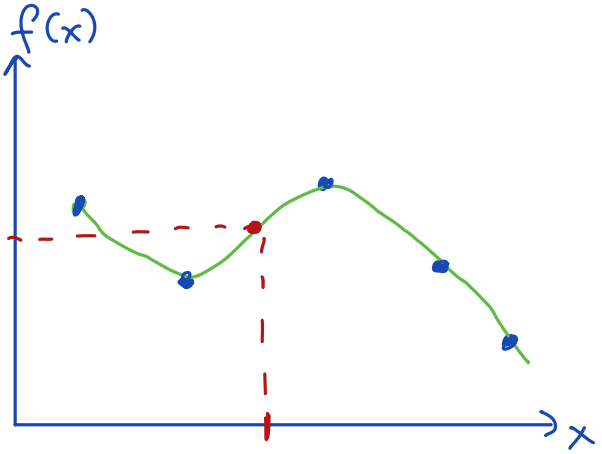
4) Curve Fitting

Regression



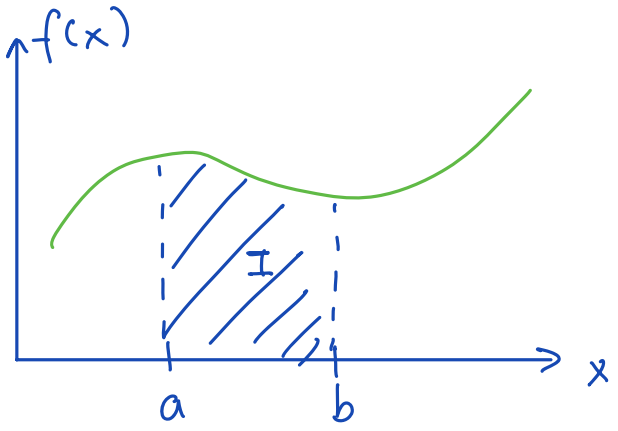
(error in data)

Interpolation



error-free data
determine intermediate values

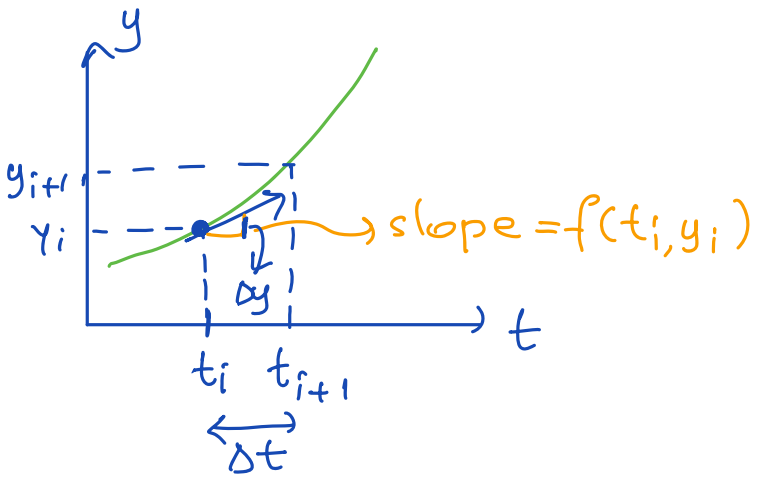
5) Integration



$$I = \int_a^b f(x)\,dx$$

Find the area under the curve

6) Ordinary Differential Equations



Given

$$\frac{dy}{dt} \cong \frac{\Delta y}{\Delta t} = f(t, y)$$

Solve for y as a function of t

$$y_{i+1} = y_i + f(t_i, y_i)\Delta t$$

Mathematical Model : a formulation or equation that expresse the essential features of a physical system or proces in mathematical tems.

Dependent variable = f (Independent variables, parameters, forcing functions)

- Dependent variable → reflects the "state" or behavior of the system
- Independent variables → time or space
- forcing functions → external influences

EX Newton's second law of motion.

$F = ma$

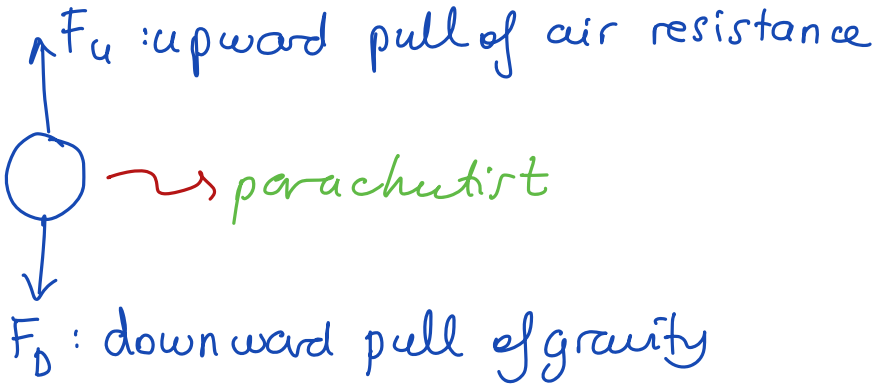
- a → acceleration (m/s^2)
- m → mass (kg)
- F → net force acting on a body (N, $kg\,m/s^2$)

$a = \frac{F}{m}$

- F → forcing function
- m → parameter
- a ↓ dependent var.

* Use Newton's second law to determine the terminal velocity of a falling body.

$$\frac{dv}{dt} = \frac{F}{m} \quad , v = \text{velocity}\ (m/s)$$



$F_D = mg$ g : acceleration due to gravity ($9.81\ m/s^2$)

$F_u = -cv$ c : drag coefficient (kg/s)
(related to the shape and surface roughness of the parachute due to air resistance)

Combining:

$$\frac{dv}{dt} = \frac{F_D + F_u}{m} = \frac{mg - cv}{m} = g - \frac{c}{m}v$$

$$\boxed{\frac{dv}{dt} = g - \frac{c}{m}v} \quad \text{differential eq}^n \quad (1)$$

Analytical Solution:

$$\boxed{v(t) = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right)} \quad (2)$$

(analytical, exact solution)

t → independent variable

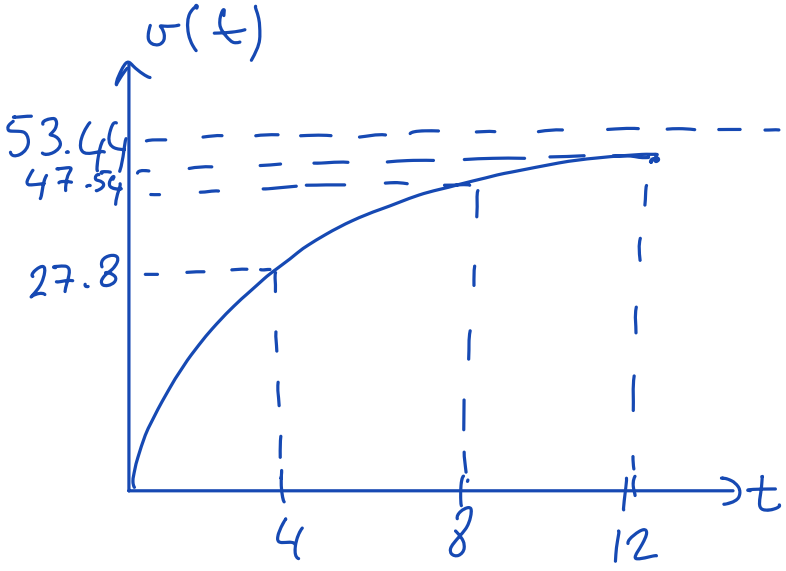
v → dependent variable

Parachutist mass : 68.1 kg
c : 12.5 kg/sec
$g = 9.81\ m/s^2$
} insert in (2)

$$v(t) = \frac{(9.81)(68.1)}{12.5}\left(1 - e^{-(12.5/68.1)t}\right)$$

$$\boxed{v(t) = 53.44\left(1 - e^{-0.183t}\right)}$$ analytical sol^n

t (sec)	v (m/s)
0	0
2	16.42
4	27.80
⋮	⋮
10	44.92
12	47.54
∞	53.44



Numerical Method: Reformulate the problem so that it can be solved by arithmetic operations.

Newton's second law of motion:

$$\frac{dv}{dt} \cong \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} \qquad (3)$$

Finite divided difference approximation of the derivative at time t_i

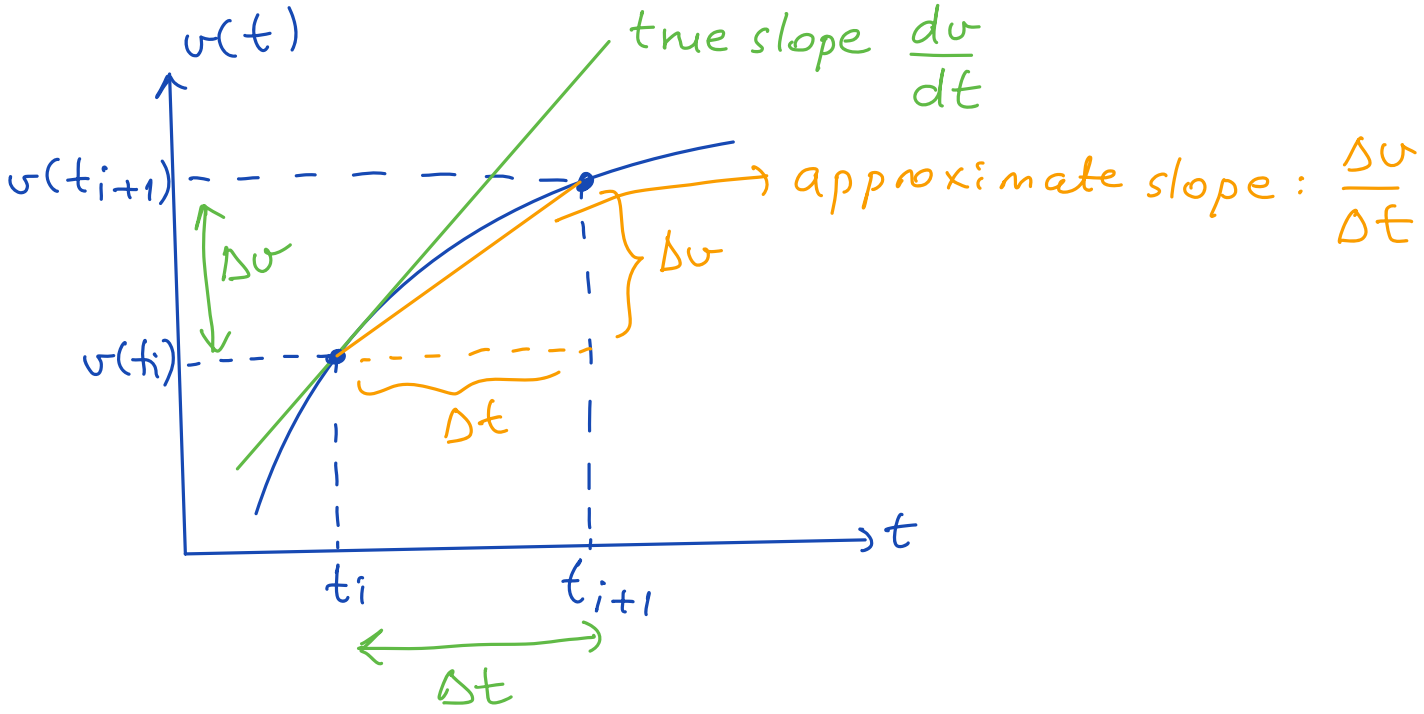
$$\frac{dv}{dt} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$$

Δv: difference in velocity

Δt: difference in time

$v(t_i)$: velocity at time t_i

$v(t_{i+1})$: " " later time t_{i+1}



Substitute (3) into (1):

$$\frac{v(t_{i+1})-v(t_i)}{t_{i+1}-t_i} = g - \frac{c}{m}v(t_i)$$

→ slope of the yellow line

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c}{m}v(t_i)\right](t_{i+1}-t_i) \qquad (4)$$

slope; step size

new value; old value; Numerical Solution

$m = 68.1\ kg$

$c = 12.5\ kg/sec$

steprize: $\Delta t = 2\ sec$

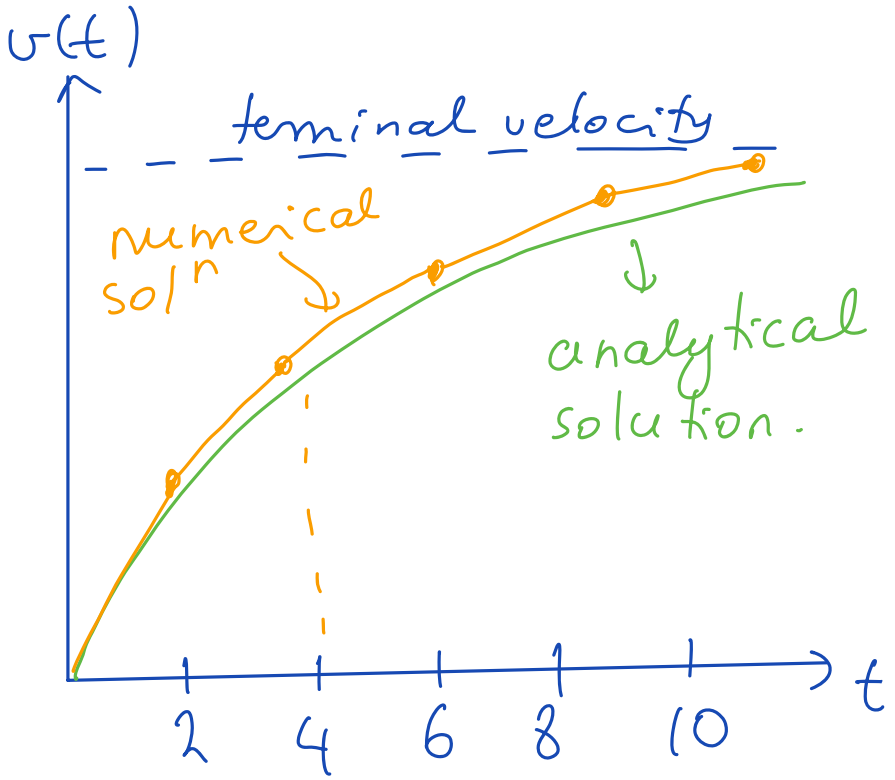
at $t_i = 0 \Rightarrow v = 0$

$$t_{i+1} = 2s \Rightarrow v = 0 + \left[9.81 - \frac{12.5}{m}\cdot 0\right]\cdot 2$$

$$v = 19.62\ m/sec$$

let $t_{i+1} = 4\,\text{sec} \Rightarrow v \dots$

t	v
0	0
2	19.62
4	32.04
⋮	⋮
10	48.02
12	50.01
∞	53.44



- Can minimize differences by using a smaller step size
- trade-off btwn. accuracy and computational effort